Prakhar Dungarwal

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EDUCATION

Columbia University — New York City, NY, USA
Master of Science in Data Science (GPA : 3.76/4) — Aug 2024 - Dec 2025

Vellore Institute of Technology — Vellore, IN
Bachelor of Technology in Computer Science and Engineering (GPA : 3.95/4) — Jul 2018 - Jul 2022

WORK EXPERIENCE

Adobe Research — New York, NY, USA
Generative AI Researcher — Aug 2025 - Present
- Developing personalized multimodal LLM frameworks using GPT-4o-v and LLava, defining 10+ evaluation factors to quantitatively assess image edits with 95% consistency across datasets.
- Building automated MLLM-as-a-Judge pipelines leveraging RAG, reasoning traces, and agentic tool binding, improving model judgment accuracy by 20% and enabling scalable, research-grade evaluation.

Intuit — Mountain View, CA, USA
AI & Data Science Intern — May 2025 - Aug 2025
- Developed an Agentic AI framework using LangGraph and ReAct-based multi-agent framework to enhance customer success metrics, reducing ground-truth annotation time by 50% via agent orchestration and internal AI tools.
- Utilized GPT-4.5 and Claude-3.5-Sonnet to extract business quality markers from call transcripts, improving recall by 30% over traditional NLP models deploying Chain-of-thought (CoT) and LLM-as-a-Judge based evaluation.

Morgan Stanley — Bengaluru, IN
Senior Associate (Data Science & Analytics) — Jan 2022 - Aug 2024
- Spearheaded development of a RAG-based chatbot operating LLaMA-3.1-8B-Instruct, all-mpnet-base-v2, and NLP techniques, cutting down query resolution time by 60% and annual ops costs by 30%.
- Improved time series forecasting by integrating CNN, Exponential Smoothing, and Prophet into a hybrid model, boosting accuracy by 20% and enabling $1M in cost savings.
- Orchestrated daily Machine Learning and Data Science operations, conducting Time Series Forecasting, refining algorithms, and implementing NLP techniques.
- Delivered business-critical data and precise predictions by training Machine Learning models, and performing critical Data Analytics tasks to manage Enterprise Data Centers, decreased annual spend by 20%.

Columbia University, Columbia Climate School, AC4 Lab — New York,NY, USA
AI & Machine Learning Researcher — Aug 2024 - Present
- Fine-tuned a RoBERTa model on GoEmotions dataset to classify YouTube videos by emotion (respect vs. contempt), integrating DeepSeek-R1 and GPT-4o models for reasoning and targeting a 30% reduction in non-peaceful watch time.
- Built and deployed a cloud-based backend for a Chrome extension used to analyse YouTube watch history using AWS (S3, EC2), LangChain, and Youtube APIs.

SKILLS & INTERESTS

- **Programming & Tools:** Python, SQL, R, C, C++, PySpark, Scala, Java, MATLAB, JavaScript, AWS, GCP, Azure, Pandas.
- **Specialized Techniques:** Machine Learning, AI Agents, Reinforcement Learning, LLMs, A/B Testing, RAG, Generative AI.
- **LLM Tools & Frameworks:** LangChain, Llamaindex, OpenAI API, HuggingFace, ChromaDB, FAISS, Streamlit, Transformer.
- **ML Frameworks:** PyTorch, TensorFlow, Scikit-Learn, Keras, Random Forest, XGBoost, MapReduce, Spark.
- **Courses:** Deep Learning for NLP, LLM Based Generative AI, Algorithms, Applied Machine Learning, Probability and Statistics.

DATA SCIENCE PROJECTS

Context-Aware Rendition with Transformers (CART) — Aug 2023
- Crafted a RAG based QA chatbot capable of understanding user context to give quick responses to operational queries leveraging advanced transformers and NLP search techniques, fine-tuned on top of Question Answering LLM's.
- Developed intelligent query search techniques by consolidating fuzzy string matching with advanced methods Cosine Similarity, enhancing chatbot response speed by 30% over traditional embedding search techniques.

Handwritten Text Recognition using Deep Learning and AI — Nov 2021
- Trained a CRNN-LSTM model on the IAM Dataset with 120,000 handwritten text samples operating PyTorch, achieving 91% accuracy and reducing paper evaluation time by 50%.
- Developed an Automated Paper Assessment tool using a CRNN model to accurately scan and transcribe handwritten scripts, outperforming conventional HMM models by 20%, with findings published in IRJET.